

Evaluating Multi-Cycle Ceramic Matrix Composites and their Aerospace Applications

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Abstract

Polymer-derived ceramic matrix composites (CMCs) are engineered ceramics designed for performance in extreme environments. Their low density, damage tolerance, high temperature strength and oxidation resistance make them attractive to the aerospace industry, specifically for rocket nozzles and jet turbines. Boron and Hafnium polysilazane liquid precursors are synthesized and used to infiltrate carbon fiber (CF) disks and bundles in a non-oxygen environment. This crosslinking process is followed by pyrolysis in an argon gas furnace. This process yields SiCN and SiBCN, depending on the precursor used during the infiltration process. The current procedure that has been reported on only has one infiltration, producing 90% ceramic yield when oxygen presence is minimized. This study experiments with the current method of fabrication for both bundles and disks to optimize the ceramic yield, material and mechanical properties of the samples by incorporating multiple infiltrations and pyrolysis cycles. FTIR analyses are used to confirm the molecular buildup and bonds created. SEM micrographs project the surface quality and coating thickness of the fibers. An oxidization study determines stability of the CMCs at temperatures up to 800 degrees Celsius.

1. Introduction

Ceramic matrix composites (CMCs) are engineered composites for extreme environments. Their low density, oxidation resistance, high damage tolerance and high temperature capability make them suitable candidates for aerospace components such as turbine blades and rocket nozzles. CMCs have about a third of the mass density and up to 200°C higher use temperatures than metal superalloys [1]. These improvements allow for elevated operation temperatures of turbines, increasing thermal efficiency. This also reduces emissions. Engine manufacturers have started incorporating CMCs into their designs, such as the CFM LEAP and the GE9x. The demand for development of these composites continues to grow as we push the capability limits of turbines. CMC capabilities would be attractive assets to your growing jet turbine research and development program at Pratt & Whitney.

Traditional ceramics are known for their high temperature capabilities, but also for their brittleness. Polymer-derived CMCs allow for the elevated temperature performance of traditional ceramics with increased strength and durability. A fabrication technique called Polymer Infiltration and Pyrolysis (PIP) produces lower volumetric yields and higher ceramic yields. In the simplest terms, CMCs are created by introducing a preceramic polymer precursor to a fiber matrix at a slightly elevated temperature. Samples are then pyrolyzed at extremely high temperatures to remove unwanted organic bonds. These two processes together are considered one cycle and turn the sample into a ceramic. Additionally, research in the past decade has shown that quaternary CMCs Si-C-X-N (X=B, Hf, Al) show exceptional stability and thermal resistance. These composites are stable at high temperatures up to 2000°C [2].

CMC technology has existed for the past 40 years and has been heavily researched by material engineers. Many different variables have been experimented upon, but the concept of multi-cycle PIP ceramic matrix composites has very little research. Introducing multiple infiltration cycles presents opportunities for improvement of porosity, purity, and strength of CMCs. This experiment aims to compare multiple cycle infiltrations of both SiCN and SiBCN ceramic matrix composites.

2. Experimentation

2.1 Materials

A commercial carbon fiber cloth (6K, 2x2 twill weave, 0.017 in. thickness, 10.9 oz/sq yd. weight) purchased from Fibre Glast Developments Corp., was chosen as the reinforcement component for this study. The polysilazane liquid precursor, Ceraset PSZ 20 (Clariant Corporation) was diluted with isopropanol ($\geq 99.9\%$, MW 60.096 g mol⁻¹, Fisher Scientific), and Trimethyl borate (TMB) ($\geq 98\%$, MW 103.91 g/mol) from Sigma-Aldrich.

2.2 Preparation of Preceramic Precursors

2.2.1. SiCN – Preceramic Polymer To achieve an acceptable viscosity to coat the CF matrix, the PSZ is diluted at a 4:3 weight ratio with the isopropanol in an inert environment. This solution is stirred at 300rpm overnight.

2.2.2 Si(B)CN – Preceramic Polymer Boron-modified silazane was prepared following procedures described in a patent[3]. In an oxygen-sealed glovebox, TMB boron reagent was added using a dropper to poly(ureamethylvinyl)silazane (commercial name: Ceraset) at a slow rate of around 1 mL/min. The ideal concentration of TMB to PSZ is around a 1:1 weight ratio. The boron-modified poly(ureamethylvinyl)silazane is then stirred at 300rpm overnight.

2.3 Preparation of PDC/CF Ceramic Mini-composites

Carbon fiber bundles are cut from the carbon fiber cloth to 5cm in length, each about 10mg in mass. 10 μ L of preceramic polymer precursors (PSZ and TMB-modified silazane) are injected into the neat carbon fiber samples using a precision pipette. The samples are then crosslinked at 180°C for 16 hours. To avoid premature organic bond formation and to achieve higher pyrolyzed and ceramic yields, the infiltration and crosslinking processes are performed in an oxygen-sealed glovebox. The samples are then pyrolyzed in a N₂ atmosphere at 800°C for 30 minutes. Figure 1 shows the high-temperature furnace program used for the pyrolysis cycles. This completes one full infiltration cycle with the resulting samples becoming ceramics. This process is then repeated, with the 1st cycle samples re-infiltrated with 10 μ L of precursor and re-pyrolyzed at 800°C. The schematic of the coating, cross-linking, and pyrolysis processes of mini-composites are presented in Figure 2.

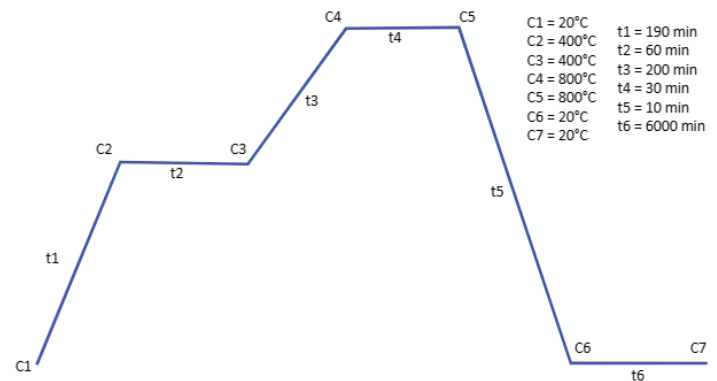


Figure 1 – Pyrolysis program diagram

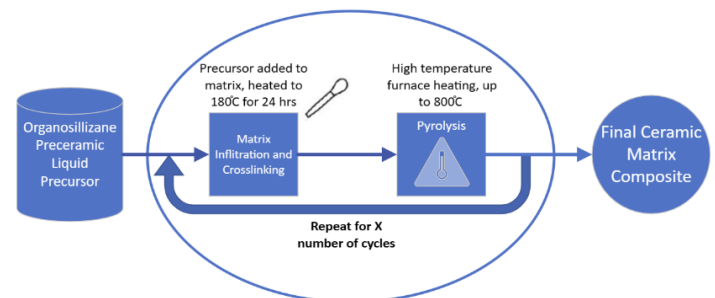


Figure 2 – PD-CMC Fabrication schematic

2.4 Characterization Techniques

This study utilized several characterization techniques to analyze the relation between cycle number and sample quality, oxidation resistance, and composition.

The surface morphology of the samples was studied using SEM on a Carl Zeiss EVO MA10 system with an incident voltage of 5 kV. The presence and evolution of various chemical functional groups of the silazane precursors in the composites were investigated using a Perkin Elmer Spectrum 400 FTIR spectrometer in the range of 500–2500 cm^{-1} . To understand the oxidation stability of the CMCs, the samples were heated to 800°C in stagnant air in a SentroTech box furnace with a heating rate of 5 °C per minute.

Oxidation testing simulates extreme environments that are applicable to aerospace components such as rocket nozzles and aircraft engines. The intense heat and presence of oxygen decomposes the integrity of the materials bonds, so it is a viable test to how the samples would perform in an aerospace environment.

3. Results and Discussion

The quality of each sample type was determined using weight analysis and visual analysis. In each intermediate stage of fabrication, every sample was weighed. These weights were averaged and used to calculate the pyrolyzed and ceramic yields using Equation 1 and Equation 2.

$$\% \text{Pyrolyzed Yield} = \frac{\text{Pyrolyzed wt.}}{\text{Crosslinked wt.}} * 100$$

Equation 1 – Pyrolyzed yield

$$\% \text{Ceramic Yield} = \frac{\text{Pyrolyzed wt.} - \text{Uncoated wt.}}{\text{Uncoated wt.}} * 100$$

Equation 2 – Ceramic yield

These metrics show the purity of the sample. Pyrolyzed yield quantifies the amount of organic matter that is burned away during the pyrolysis

stage, and ceramic yield quantifies the amount of ceramic created by the entire process. The experimental yield results are compiled in Table 1.

Type	Cycle	Pyrolyzed Yield	Ceramic Yield
SiCN	1	83.94%	76.17%
	2	87.07%	31.32%
	3	92.51%	26.69%
SiBCN	1	95.98%	25.02%
	2	94.56%	18.75%
	3	96.46%	11.71%

Table 1 – Pyrolyzed and Ceramic yields on SiCN/SiBCN samples

When analyzing these results, it is clear that there is a relatively steady increase in pyrolysis yield as the number of cycles increases. However, ceramic yield decreases with the increasing number of cycles. This is due to the fact that the baseline weight of the sample increases with each cycle, and there is less surface area available for ceramic coating to form. The organic bonds are still dissipating at a very high rate, shown by the pyrolyzed yield, which is desirable. The fact that ceramic yield is still greater than 10% at three cycles shows that there is still a significant amount of ceramic formation which increases the integrity of the material.

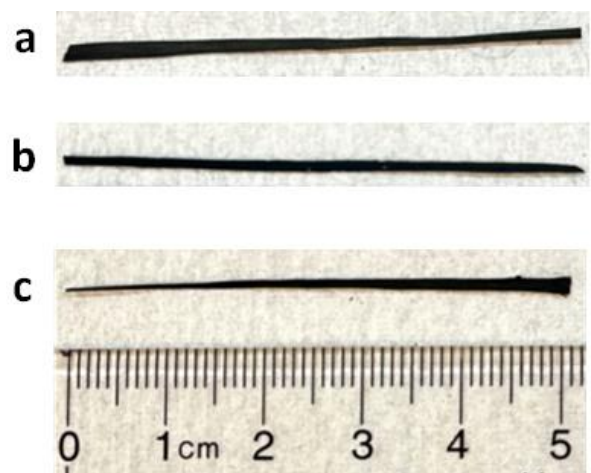


Figure 3 – a.) Neat carbon fiber, b.) Crosslinked sample, c.) Pyrolyzed sample

SEM micrographs of the CF/SiCN and CF/SiBCN samples revealed information about the ceramic coating of the individual fibers of each different

sample type. Images were taken at 800x and 1500x magnification of each sample type, as shown in Figure 4 and Figure 5. Both CF/SiCN and CF/SiBCN ceramic coverage are rough and shard-like. In Figures 4a, 4b, and 4d, there is a clear distinction between ceramic and exposed carbon fiber. This can be attributed to uneven application of the precursor during the infiltration phase. SiCN Cycle 3, SiBCN Cycle 2, and SiBCN Cycle 3 show excellent ceramic coverage. However, there is excess ceramic on the surface. The highest quality of ceramic is SiBCN Cycle 3, which has smooth, uniform coverage on each individual carbon fiber strand, as shown in both the 800x and 1500x magnified images (Figure 4f and Figure 5f). There is a striking difference between the SiCN and SiBCN Cycle 3 samples, which is most likely due to the difference in viscosity of the precursors.

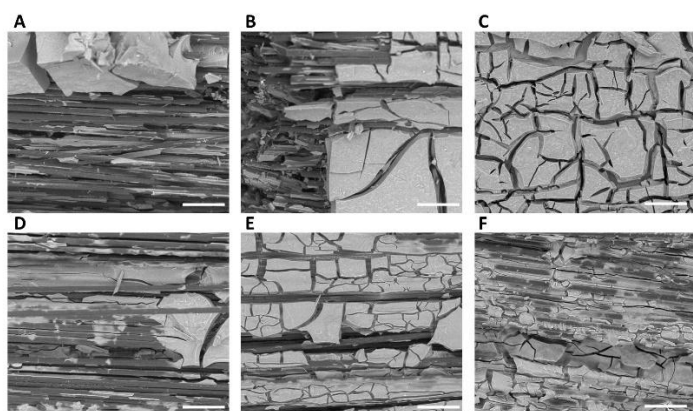


Figure 4 – 1500x magnification, scale bar has a length of 50 μm . a.) SiCN Cycle 1, b.) SiCN Cycle 2, c.) SiCN Cycle 3, d.) SiBCN Cycle 1, e.) SiBCN Cycle 2, f.) SiBCN Cycle 3.

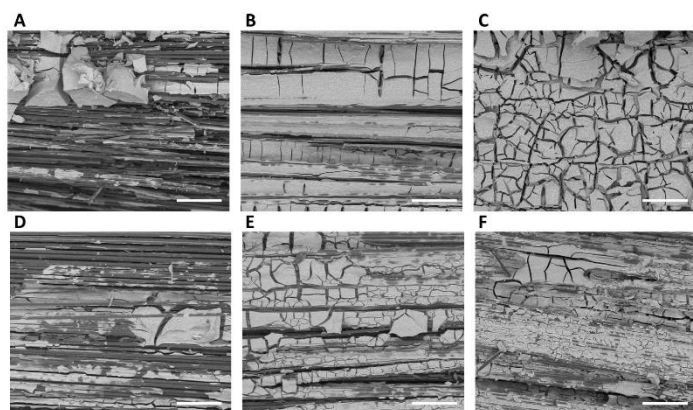


Figure 5 – 800x magnification, scale bar has a length of 100 μm . a.) SiCN Cycle 1, b.) SiCN Cycle 2, c.) SiCN Cycle 3, d.) SiBCN Cycle 1, e.) SiBCN Cycle 2, f.) SiBCN Cycle 3

3.1 Characterization

FTIR analysis of the SiCN and SiBCN samples confirm the samples' conversion from polymer to ceramic. The FTIR spectra were compared between each cycle, but it was found that there was very little variation between the plots. This is expected since the chemical composition of the sample should not change with the amount of precursor applied and number of pyrolysis cycles. For this reason, only Cycle 1 SiCN and SiBCN FTIR Spectra are shown in Figure 6 and Figure 7 respectively.

In Figures 6 and 7, the peaks at 1260 and 1060 cm^{-1} corresponded to Si-CH₃ and Si-O-Si respectively. These peaks confirm the presence of silicon in the polymer precursor. Additionally, the absorption band observed at 1175 cm^{-1} corresponded to the Si₂-NH. For Si(B)CN/CF, the decreased peak intensities of Si-CH₃ and Si₂N-H were contributed to the improved crosslinking of the samples at 180°C. A wide peak of the B-N functional group formed in the crosslinking stage and became more intense after pyrolysis (Figure 7). For both precursor types, the CMC samples contained two large peaks of Si-O (1060 cm^{-1}) and Si-C (800 cm^{-1}) after pyrolysis. The presence of these two obvious peaks in all the samples confirmed the existence of ceramic coatings in the composites. All characteristic absorbance bands of the H-containing functional groups, such as for Si-CH₃ (1265 cm^{-1}), C-H (1370 cm^{-1}), and Si-H (2165 cm^{-1}), had all clearly diminished. This FTIR spectra analysis confirmed the successful conversion from of the polymer precursor to a ceramic material.

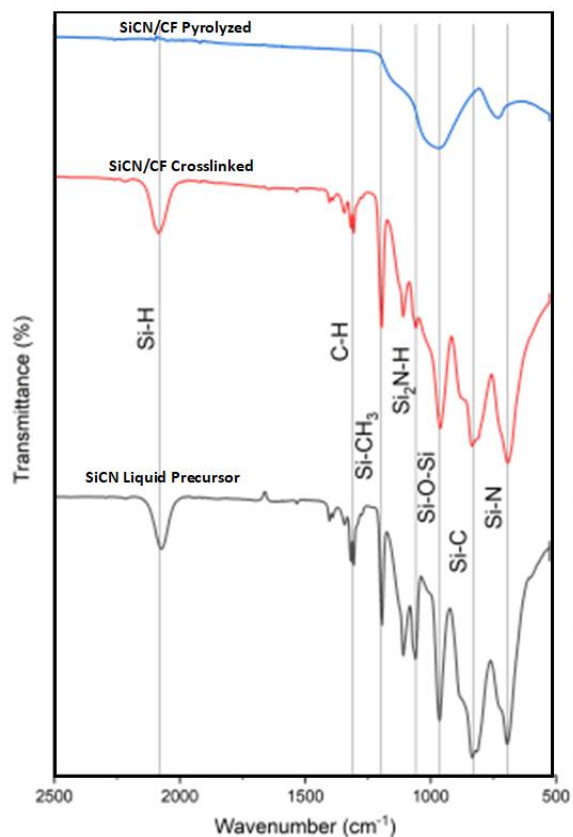


Figure 6 – FTIR Spectra of PSZ Precursor, SiCN crosslinked sample, and SiCN Pyrolyzed sample

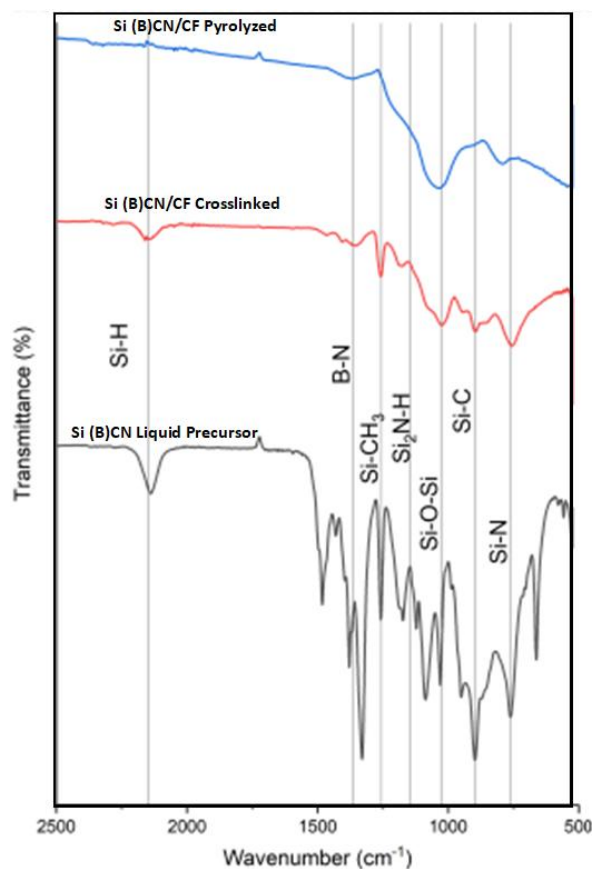


Figure 7 – FTIR Spectra of PSZ-TMB Precursor, SiBCN crosslinked sample, and SiBCN Pyrolyzed sample

3.2 Oxidation

The oxidation behaviors of the SiCN and SiBCN samples at 800°C are shown in Figure 8 and Table 2. Six CMC samples were oxidized in an inert environment for 8 hours. An uncoated carbon fiber sample is also tested as a control. The results show that all ceramic samples survive the oxidation test, while the carbon fiber completely disintegrates. Figure 8 shows the visual results and discoloration of the samples. SiCN Cycle 1, SiBCN Cycle 1, 2, and 3 all turned white on the surface, indicating the formation of SiO_2 . SiCN and Cycle 2 and 3 show excellent resistance to surface oxidation. Each sample was weighed before and after the test, and the weight retention percentage was calculated and reported in Table 2. Overall, SiCN shows the highest weight retention with SiCN Cycle 3 having the largest of all samples at 69.71%. However, for both precursor types, weight retention and number of cycles have a direct relationship.

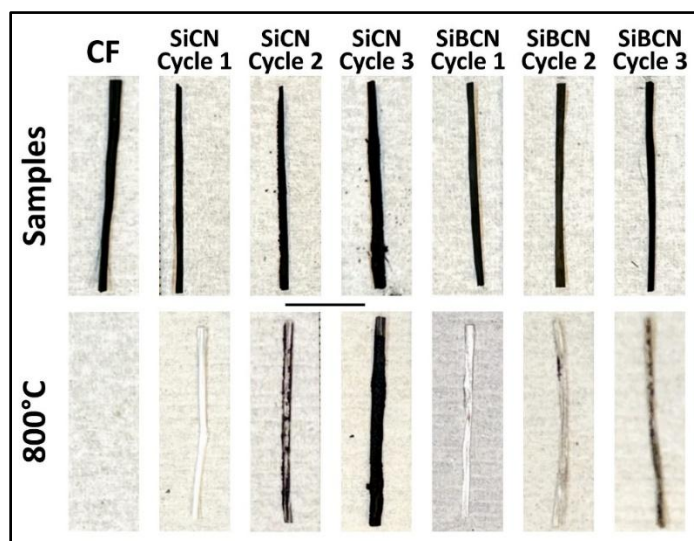


Figure 8 – Oxidation test results

Type	Cycle	Weight retention (%) at 800°C
SiCN	1	27.03%
	2	48.21%
	3	69.71%
SiBCN	1	18.33%
	2	29.33%
	3	36.05%

Table 2 – Oxidation weight retention

4. Conclusion

The results of the experiment conclusively prove that increased number of infiltration cycles has a positive impact on CMC quality. When comparing yields, consistent increase in pyrolyzed yields confirms the dissipation of organic bonds, and the presence of ceramic yields greater than 10% shows the consistent creation of ceramic coverage on the carbon fiber strands. FTIR analysis confirms the polymer to ceramic transformation. SEM imaging

shows the ceramic coverage that occurs for all sample types.

SiCN and SiBCN Cycle 3 performed the best in comparison to other samples, confirming the continued research and pursuit of multi cycle ceramic matrix composites is worth the investment. Future experiments may include different polymer precursors such as SiHfCN, tensile testing, oxidation testing at higher temperatures, and SEM analysis of oxidized samples.

5. References

- [1] Romil Bhandavat and G. Singh, "Synthesis, Characterization, and High Temperature Stability of Si(B)CN-Coated Carbon Nanotubes Using a Boron-Modified Poly(ureamethylvinyl)Silazane Chemistry," *Journal of the American Ceramic Society*, vol. 95, no. 5, pp. 1536–1543, Feb. 2012, doi: <https://doi.org/10.1111/j.1551-2916.2012.05079.x>.
- [2] Shakir Bin Mujib, M. Rasheed, and G. Singh, "Evaluating Use of Boron- and Hafnium-Modified Polysilazanes for Ceramic Matrix Minicomposites," *ACS Omega*, vol. 7, no. 49, pp. 45325–45335, Dec. 2022, doi: <https://doi.org/10.1021/acsomega.2c05916>.
- [3] Singh, G.; Bhandavat, R. Boron-modified silazanes for synthesis of SiBNC ceramics. U.S. Patent 20,150,030,856 A1, 2016.
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- [5] Mujib, Shakir, et al. Hybrid HfC-SiCN Matrix for Improved Oxidation Resistance of Carbon Fiber Reinforced Mini-Composites. *Ceramic Engineering & Science*, 8 Feb. 2024.